JARRED M. CARTER
New York, NY | jarredmcarter.com

EDUCATION

M.S., Cybersecurity, New York University, May 2025, GPA 3.8/4.0
- Coursework: Information Security and Privacy, Network Security, Pentesting and Vulnerability Analysis, Application Security, Computer Networking, Digital Forensics, Reverse Engineering, Web Application Exploitation
- Certifications: CompTIA Security+, OSCP+ (in progress, training on Hack The Box)
- CyberCorps Scholarship for Service Scholar, awarded to top 1% of the class for two years in a row.

B.S., Computer Science & B.S., Cybersecurity, Marshall University, Dec. 2022, GPA 3.7/4.0

RELEVANT EXPERIENCE

Security Analyst 06/2024 — 09/2024
Open Law Library, Washington, D.C.
- Aggregated and analyzed vulnerability and system data to support risk assessment, reporting, and prioritization across application and infrastructure environments.
- Developed documentation and reporting workflows to improve adoption of security tooling and remediation processes by engineering and operations teams.
- Supported 40k+ users by maintaining operational security tooling to ensure continuous verification of code commits, preventing malicious activity on public law hosting infrastructure.
- Supported the vulnerability management program by validating findings, tracking remediation progress, and ensuring adherence to internal triage and remediation SLAs.

Penetration Testing Engineer 08/2022 — 12/2022
City National Bank, Cross Lanes, WV
- Maintained and operated vulnerability scanning (Nessus), detection, and monitoring systems supporting continuous identification of security risks.
- Provided technical guidance to SOC analysts and infrastructure teams on vulnerability assessment methodology and security tooling, supporting knowledge transfer and team capability building.
- Conducted adversarial testing and vulnerability assessments across enterprise infrastructure, validating findings and translating technical results into risk-based remediation recommendations for technical and executive stakeholders.
- Collaborated with SOC analysts, infrastructure, and IT operations teams to prioritize vulnerabilities based on risk, exploitability, and asset criticality, serving as technical liaison between security and business units.
- Performed incident response and threat analysis including malware payload analysis, email investigation, and network traffic analysis using Wireshark and security tooling to identify anomalous activity and TTPs aligned with MITRE ATT&CK framework.

Information Systems Specialist I 05/2018 — 08/2023
WorkForce West Virginia, Charleston, WV
- Increased cross-functional visibility by 70% across 18 centers statewide through design, implementation, and maintenance of live dashboards and KPIs for IT asset inventory, system usage, and lifecycle tracking.
- Communicated cross-functional IT strategy to executive leadership, presenting data-driven recommendations for infrastructure investment and asset lifecycle decisions.
- Cut ~$150k in costs by designing and deploying in-house systems that streamline software licensing and hardware usage tracking across 18 career centers, improving IT asset visibility, and reducing waste.
- Managed IT asset lifecycle (provisioning, de-provisioning, end-of-life) and periodic user access reviews for OSCAR, RAPIDS, DMV, and related state systems, maintaining SOPs and access documentation for federal and state program compliance.

SKILLS

Languages: English (native), Spanish, Python, SQL, HTML, CSS, Flask, C#, .NET, Swift, SwiftUI (iOS/macOS).
GRC & Compliance: Risk assessment methodologies, control implementation and testing, audit evidence collection, access control frameworks (least privilege, SoD, privileged access management), compliance documentation.
Communication: Communication for technical and non-technical stakeholders, collaboration, conflict resolution, training, attention to detail, client outreach.
Cloud & Infrastructure: AWS, Azure, GCP (coursework-level deployment and testing), Docker, Kubernetes (container security and orchestration through coursework and projects)
Pentesting & Vulnerability Assessment: Metasploit, Nmap, Recon-ng, AMASS, Maltego, Shodan, FOCA, Nessus, Scalpel, FTK Imager, Autopsy, GDB, Binary Ninja/Ghidra, Burp Suite, IDA Pro, Django Security, Container Security, Mobile App Security.
Vulnerability Management & AppSec: Triage and remediation tracking, application security testing, false-positive reduction.
Incident Response & Threat Analysis: Malware payload analysis, email/header investigation, network traffic analysis (Wireshark), anomaly detection, TTP identification and MITRE ATT&CK mapping.

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PROJECTS

Pentesting and Vulnerability Analysis Project Fall 2023

- Conducted comprehensive penetration testing for NBN Corporation, identifying critical vulnerabilities including Cross-Site Scripting (XSS), SQL Injection, Local File Inclusion, and unauthorized server access, leading to root access on all systems.
- Assessed and scored vulnerabilities using NIST's CVSS Version 3, with an average criticality score of 8.86, highlighting the high risk and immediate need for mitigation.
- Recommended immediate security fixes and mitigations, including sanitization of user input, system updates, enhanced password policies, and implementing microservice architecture to isolate network assets.

Digital Forensics Final Project Spring 2024

- Investigated email-based attack chains by analyzing malicious attachments and extracting indicators of compromise to map threat actor delivery techniques.
- Investigated phishing and BEC attack vectors, email headers, attachments, and links to determine maliciousness/attack patterns.
- Reassembled deleted and obfuscated evidence from 8 system images using Scalpel and FTK Imager for recovery of hashes, user accounts, logs, documents, and metadata to deepen digital evidence recovery insights.
- Identified 80+ functions, libraries, and security vulnerabilities by reverse engineering unknown executables, creating and inspecting forensic images/file systems with Procmon and PeStudio.

Threat Intelligence Investigation: Automated Bot Traffic & Ad Fraud Analysis Spring 2026

- Analyzed 126,959 e-commerce session records with Python and SQL, identifying 25.51% as automated bot traffic through anomaly detection on interaction timing, source infrastructure (ASN), and device fingerprint reuse.
- Correlated indicators across network, device, and application data, including datacenter ASNs, headless browser signatures, timezone mismatches, and inhuman interaction speeds, to separate malicious automation from legitimate users.
- Quantified $3,362–$13,448 in wasted ad spend by tracing bot activity to checkout, login, and product endpoints, revealing patterns consistent with credential stuffing and inventory hoarding.
- Developed a prioritized mitigation strategy covering ASN blocking, device fingerprint detection, checkout rate limiting, and MFA enforcement.

Retrac Labs: Privacy-First iOS & macOS Apps (Founder) 2025 — Present

- Founded a studio that has shipped privacy-first Apple-platform apps, built with Claude Code (agentic AI coding). No accounts, analytics, or servers; the macOS apps ship without the network entitlement.
- Hash Drop (macOS, in development): File-integrity verifier that checks hashes, PGP/minisign/SSH-signed checksum lists, code signatures, and Apple notarization, and flags files disguised as another type, all offline.

Offensive Security CTF Challenges Fall 2024

- Analyzed malicious artifacts and reverse engineered 20+ malware samples using Ghidra and Binary Ninja to identify threat actor TTPs and attack chains.
- Conducted root cause analysis of 20+ security vulnerabilities across memory corruption and application security issues, transforming technical findings into actionable remediation strategies using reverse engineering methodologies and enterprise-grade analysis tools.

Application Security Engineering Projects Spring 2024

- **AI-Ready Security Assessment:** Identified and remediated 4 critical vulnerabilities (XSS, CSRF, SQL Injection, command injection) in Django web application; implemented automated detection rules and established security controls applicable to AI/ML API environments.

AWARDS & PUBLICATIONS

- J.M. Carter, Y. Makula, D. Sheshappa, S.P. Virigineni, Lightweight Threat Model for Emissary Ingress, Cloud Native Computing Foundation (2026).
- J.M. Carter, C.E. Morris, M.J. Oliver, Addition of Stahl Heater to the Propulsion System Laboratory Simulator’s 450lb Airline, NASA Internal Paper (2021).